

How to Bake a Cake – Beginner's Guide

Lesson 2: Essential Ingredients & Measurements

Essential Ingredients & Measurements – Study Notes

Key Concepts

1. **Basic pantry staples** – flour, sugar, eggs, butter, and leavening agents.
2. **Function of each ingredient** – how they affect texture, structure, flavor, and moisture.
3. **Standard US volume measurements** – cups, teaspoons, tablespoons.
4. **Weight conversions** – why grams are more accurate and the typical gram equivalents for common ingredients.
5. **Practical conversion tip** – using a digital kitchen scale and a quick reference chart.

Summary

This lesson introduces the five cornerstone ingredients found in most baked goods: **flour**, **sugar**, **eggs**, **butter**, and **leaveners** (baking powder, baking soda, and yeast). Understanding the role each plays helps you troubleshoot recipes and create your own.

In addition, the lesson covers the crucial skill of converting **cup measurements to grams**. While cups are convenient, they can vary by ingredient density and by how the cup is filled. We provide standard gram equivalents for the core ingredients and a simple method for accurate conversion using a kitchen scale. Mastering these basics gives you a solid foundation for consistent, repeatable results in the kitchen.

Important Points

- **Flour** – provides structure; most recipes call for all purpose flour (120 g per cup).
- **Sugar** – adds sweetness, contributes to browning and tenderness; granulated sugar (200 g per cup).
- **Eggs** – act as binders, leaveners, and moisture carriers; one large egg (50 g (1/4 cup)).
- **Butter** – adds richness, flavor, and tenderness; 1 cup (2 sticks) (227 g).
- **Leaveners** – create gas bubbles for rise:
 - Baking powder: 1 tsp (4 g)
 - Baking soda: 1 tsp (5 g)
 - Yeast (active dry): 1 tsp (3 g)
 - **Why weigh?**
 - Volume can be inconsistent (e.g., packed brown sugar vs. loosely scooped).
 - Weight gives repeatable results, especially for delicate pastries.
 - **Conversion cheat sheet (US cup to grams):**

| Ingredient | 1/2 cup | 1 cup | 2 cups |

	60 /g	120 /g	240 /g
All purpose flour	60 /g	120 /g	240 /g
Whole wheat flour	55 /g	110 /g	220 /g
Granulated sugar	100 /g	200 /g	400 /g
Brown sugar (packed)	110 /g	220 /g	440 /g
Powdered sugar	60 /g	120 /g	240 /g
Butter (softened)	113 /g	227 /g	454 /g
Milk / water	120 /g	240 /g	480 /g
Honey / maple syrup	170 /g	340 /g	680 /g

• **Conversion tip:** Place your empty measuring cup on the scale, tare to zero, then add the ingredient until the desired gram weight is reached.

Examples

Example 1 – Converting a Simple Pancake Recipe

Original (US cups):

- 1 1/2 /cups all purpose flour
- 2 /Tbsp granulated sugar
- 1 /tsp baking powder
- 1/4 /tsp salt
- 1 /cup milk
- 1 /large egg
- 2 /Tbsp melted butter

Converted to grams:

Ingredient	Cups/Tbsp/Tsp	Grams
Flour	1.5 /cups	180 /g
Sugar	2 /Tbsp ("H /S /cup)	67 /g
Baking powder	1 /tsp	4 /g
Salt	1/4 /tsp	1.5 /g
Milk	1 /cup	240 /g
Egg	1 large	50 /g
Butter	2 /Tbsp (1/4 /cup)	57 /g

Explanation: Using the gram chart eliminates the guesswork of “scooping” flour and ensures the batter’s consistency is the same every time.

Example 2 – Adjusting a Cookie Recipe for a Scale Only Kitchen

Original (cups): 2 /cups flour, 1 /cup butter, 1 /cup brown sugar, 1/2 /cup granulated sugar, 2 /eggs, 1 /tsp baking soda.

Step by step conversion:

1. Weigh each ingredient using the cheat sheet:

- Flour: 2 /cups !' 240 /g
- Butter: 1 /cup !' 227 /g

- Brown sugar (packed): 1 /cup !' 220 /g
- Granulated sugar: ½ /cup !' 100 /g
- Eggs: 2 /× /50 /g !' 100 /g (or crack and weigh the shells together, then subtract).
- Baking soda: 1 /tsp !' 5 /g

2. Mix according to the original method; the texture will match the intended result because the weight ratios are preserved.

Quick Review

1. What structural role does flour play in baked goods?
2. How many grams are in a standard US cup of granulated sugar?
3. Why is weighing butter more reliable than measuring it by volume?
4. Convert ¾ /cup of all purpose flour to grams.
5. Which leavening agent would you use for a quick rise biscuit: baking powder or baking soda, and why?

Further Reading

- ****Ingredient Functions in Baking**** – deeper dive into gluten development, creaming butter & sugar, and egg emulsification.
 - ****Understanding Different Flours**** – cake flour, bread flour, whole grain flours, and their protein percentages.
 - ****Advanced Measurement Techniques**** – using a digital scale with tare function, measuring sticky ingredients (honey, syrup), and temperature considerations.
 - ****Leavening Chemistry**** – how acids, bases, and moisture interact in baking powder vs. baking soda vs. yeast.
- Happy baking! Ø<ß,

